

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-01 to 03

Description	Specified	Offered
Numbers of units	3	3
Type	Fresh Water-cooled, Centrifugal Chiller	Water-Cooled Centrifugal Chiller
Manufacturer	-	YORK
Country of origin	-	PRC
Model Number	-	YKIBKRH95CVG
Cooling Capacity, (kW)	3517	3517
Refrigerant used	HFC	R-134a
Total Input Power, (kW)	-	610.2
COP	5.7	5.76
NPLV (according to AHRI 550/590)	≥ 9.0	10
(1) COMPRESSOR & MOTOR		
Type of compressor	Scroll / Screw / Centrifugal	Centrifugal
Number of compressor	-	1
Compressor speed, (rpm)	<12000-	< 12000
Capacity control type	Inlet Guide Vanne/ VSD	IGV + VSD
Minimum Part Load (%)	-	15%
Motor rating (kW)	-	589 (Overall)
Type of compressor motor starter	Variable Speed Drive	VSD
Full-Load Current, (A)	-	1024
Starting current, (A)	-	0 to Full-load
Type of compressor motor	-	Open Type
Overall equipment power factor	0.85	>0.9
Type of drive	-	Gear

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-01 to 03

Description	Specified	Offered
(2) EVAPORATOR		
Type	Shell and Tube	Shell & Tube
Tube material	Copper	Copper
Tube sheet material	steel-	Steel
No. of passes	-	2
Chilled water entering temp, (°C)	12.5	12.5
Chilled water leaving temp, (°C)	7	7
Chilled water flow rate (L/s)	-	152.5
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	72.5
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	0.018	0.018
Insulating material	-	Closed Cell Foam
Thickness of insulation (mm)	--	38mm
Water box	Required	Marine Water Box

(3) CONDENSER		
Type	Shell and Tube	Shell & Tube
Heat rejection capacity (kW)	-	4099
Tube material	Copper	Copper
Tube sheet material	Steel	Steel
No. of passes	-	2
Entering water temperature (°C).	32	32
Leaving water Temperature (°C).	37	37
Water flow rate (L/s)	-	196.2
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	93.6
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	0.044	0.044
Water box	Required	Marine Water Box

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-01 to 03

Description	Specified				Offered			
(4) MISCELLANEOUS								
Operating weight per unit (kg)	-				20296			
Octave Band Center Frequency, (Hz)	63	125	250	500	1k	2k	4k	8k
Sound Pressure level at 1 m distance, (dB)	75.5	80	81	80.5	76.5	80	72.5	68
Overall sound pressure level at 1 m distance, (dBA)	-				84			
Overall dimensions (L x W x H), (mm)	-				5198 x 2523 x 3300			
Volts/Phase/Hertz	380V / 3ph / 50Hz				380 / 3 / 50			
Part-Load Performance								
NPLV (according to AHRI 550/590)	≥ 9.0				10			
100%	5.7				5.76			
75%	-				8.15			
50%	-				11.93			
25%	-				9.74			

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-04

Description	Specified	Offered	
		Cooling Mode	HR Mode
MODE			
Numbers of units	1	1	
Type	Fresh Water-cooled, Heat Recovery Chiller	Water-Cooled Heat Recovery Chiller	
Manufacturer	-	YORK	
Country of origin	-	PRC	
Model Number	-	YKKSOXH95CWG	
Cooling capacity, (kW)	-	3517	2902
Heating Capacity, (kW)	-	N/A	3549
Refrigerant used	HFC	R-134a	
Total Input Power, (kW)	-	614.8	686.8
Cooling COP	5.7	5.7	N/A
Heating + Cooling COP	-	N/A	9.39
NPLV (according to AHRI 550/590)	≥ 9.0	9.88	N/A
(1) COMPRESSOR & MOTOR			
Type of compressor	Scroll / Screw / Centrifugal	Centrifugal	
Number of compressor	-	1	
Compressor speed, (rpm)	-	< 12,000	
Capacity control type	-	IGV + VSD	
Minimum Part Load (%)	-	15%	
Motor rating (kW)	-	672	
Type of compressor motor starter	Variable Speed Drive	VSD	
Full-Load Current, (A)	-	1142	
Starting current, (A)	-	0 to full load	
Type of compressor motor	-	Open Type	
Overall equipment power factor	0.85	> 0.9	
Type of drive	-	Gear	

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-04

Description	Specified	Offered	
MODE		Cooling Mode	HR Mode
(2) EVAPORATOR			
Type	Shell and Tube	Shell & Tube	
Tube material	Copper	Copper	
Tube sheet material	-	Steel	
No. of passes	-	2	
Chilled water entering temp, (°C)	12.5	12.5	
Chilled water leaving temp, (°C)	7	7	
Chilled water flow rate (L/s)	-	152.7	
Working pressure (kPa)	-	1600	
Water pressure drop (kPa)	-	52.5	
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	-	0.018	
Insulating material	-	Closed Cell Foam	
Thickness of insulation (mm)	-	38	
Water box	-	Marine Water Box	

(3) DOUBLE-BUNDLE CONDENSER			
		CT Condenser	HR Condenser
Type	Shell and Tube	Shell & Tube	Shell & Tube
Heat rejection capacity (kW)	-	4117	N/A
Heating Capacity, (kW)	-	N/A	3549
Tube material	Copper	Copper	Copper
Tube sheet material	-	Steel	Steel
No. of passes	-	2	2
Entering water temperature (°C).	32	32	40.5
Leaving water Temperature (°C).	37	37	45.5
Water flow rate (L/s)	-	194	172
Min. Condenser Water Flow Rate (L/s)	-	N/A	N/A
Working pressure (kPa)	-	1600	1600
Water pressure drop (kPa)	-	58.5	91.1
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	-	0.044	0.018
Water box	-	Compact Water box	Compact Water box

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-04

Description	Specified				Offered			
MODE					Cooling Mode	HR Mode		
(4) MISCELLANEOUS								
Operating weight per unit (kg)	-				30797			
Octave Band Center Frequency, (Hz)	63	125	250	500	1k	2k	4k	8k
Sound Pressure level at 1 m distance, (dB)	75.5	80.5	81.5	80.5	76.5	79.5	72.5	68
Overall sound pressure level at 1 m distance, (dBA)	-				84			
Overall dimensions (L x W x H), (mm)	-				5266 x 2888 x 3502			
Volts/Phase/Hertz	380V / 3ph / 50Hz				380 / 3 / 50			
Part-Load Performance								
NPLV (according to AHRI 550/590)	≥ 9.0				9.88	N/A		
100%	5.7				5.72	N/A		
75%	-				8.08	N/A		
50%	-				11.75	N/A		
25%	-				9.57	N/A		

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-05 to 06

Description	Specified	Offered
Numbers of units	-	2
Type	Fresh Water-cooled, Oil Free Chiller	Water-Cooled Oil-Free Chiller
Manufacturer	-	YORK
Country of origin	-	PRC
Model Number	-	YMC2-S1759AB
Cooling Capacity, (kW)	-	1759
Refrigerant used	-	R-134a
Total Input Power, (kW)	-	308
COP	≥ 5.70	5.71
NPLV (according to AHRI 550/590)	≥ 9.0	10.8
(1) COMPRESSOR & MOTOR		
Type of compressor	Reciprocating/Scroll/ Centrifugal/Screw	Magnetic Bearing Centrifugal
Number of compressor	-	1
Compressor speed, (rpm)	≤ 48000	< 20000
Capacity control type	Variable Speed Drive	VSD
Minimum Part Load (%)	-	15%
Motor rating (kW)	-	308 (overall)
Type of compressor motor starter	Variable Speed Drive	VSD
Full-Load Current, (A)	-	493
Starting current, (A)	-	0 to Full load
Type of compressor motor	Hermetic / Open-	Semi-Hermetic
Overall equipment power factor	0.85	> 0.9
Type of drive	-	Direct Drive

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-05 to 06

Description	Specified	Offered
(2) EVAPORATOR		
Type	Shell & Tube	Shell & Tube
Tube material	-	Copper
Tube sheet material	-	Steel
No. of passes	-	2
Chilled water entering temp, (°C)	12.5°C	12.5
Chilled water leaving temp, (°C)	7°C	7
Chilled water flow rate (L/s)	-	76.2
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	29.3
Fouling factor ($\text{m}^2 \text{ } ^\circ\text{C}/\text{kW}$)	0.018	0.018
Insulating material	-	Closed Cell Foam
Thickness of insulation (mm)	-	38mm
Water box	-	Marine Water Box

(3) CONDENSER		
Type	Shell and Tube	Shell & Tube
Heat rejection capacity (kW)	-	2093
Tube material	-	Copper
Tube sheet material	-	Steel
No. of passes	-	2
Entering water temperature (°C).	32	32
Leaving water Temperature (°C).	37	37
Water flow rate (L/s)	-	100
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	36
Fouling factor ($\text{m}^2 \text{ } ^\circ\text{C}/\text{kW}$)	0.044	0.044
Water box	-	Marine Water Box

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-05 to 06

Description	Specified				Offered			
(4) MISCELLANEOUS								
Operating weight per unit (kg)	-				10282			
Octave Band Center Frequency, (Hz)	-	-	-	-	1k	2k	4k	8k
Sound Pressure level at 1 m distance, (dB)	-	-	-	-	66	68	66	66
Overall sound pressure level at 1 m distance, (dBA)	-				73			
Overall dimensions (L x W x H), (mm)	-				3861 x 1892 x 2486			
Volts/Phase/Hertz	-				380 / 3 / 50			
Part-Load Performance								
NPLV (according to AHRI 550/590)	≥ 9.0				10.8			
100%	5.7				5.71			
75%	-				8.24			
50%	-				13.18			
25%	-				11.92			